CEG4158 - Computer Control in Robotics

Robotics Laboratory Guide

Lab session #2

Robotic Manipulator Forward Kinematics

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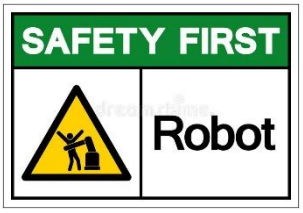
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WARNING! Always play safe with robots.

1 Introduction

When developing the forward kinematic model for a commercial robotic platform, it is critical to closely study and understand the specifications from the manufacturer, as well as pay attention to the physical movements of the machine, in order to derive the model properly. This laboratory session builds upon your previous experience with the Kinova Gen3 lite manipulator in Lab. #1, and on your attentive examination of the relevant documentation in the manufacturer's User Guide "*Gen3_lite_USER_GUIDE_release2021.pdf*" available on Brightspace, under Labs. Take the time to study all the relevant information from the documentation. If you derive a wrong model from the beginning, it will have major impacts on developing a controller for the robot.

2 Objectives

This second laboratory session aims to provide you with an opportunity to develop a forward kinematic model while interacting with a real robot rather than only a static schematic representation of it. The fact that you can interact with the robot gives you the possibility to understand how the kinematic chain, formed of a sequence of joints and rigid members, makes the robot's end effector move in 3D space. The specific objectives include:

1. Use the CoppeliaSim simulation of the Kinova Gen3 lite robot with 6 degrees-of-freedom to study its mechanical movements;
2. Apply the Denavit-Hartenberg convention to model the Gen3 lite manipulator;
3. Derive the corresponding *A* matrices for the robot;
4. Form the forward kinematic transformation matrix that represents the relationship from the base to the end effector of the manipulator, in accordance with the specifications provided by the manufacturer;
5. Use the CoppeliaSim simulator to experimentally validate the mathematical correctness of your forward kinematic model;
6. Expose participants to the challenges encountered when modeling a robotic manipulator;
7. Expand your familiarity with programming functionalities used on a commercial robotic platform.

3 Experimental Protocol

Follow the steps below to develop your own forward kinematic model of the Kinova Gen3 lite manipulator with 6 revolute degrees of freedom.

3.1 Getting started

1. Follow the procedures presented in Lab. #1 and connect to one of the STE-3040 workstations.
2. Execute the necessary configuration adjustments as described in the Laboratory Guide for session #1.
3. Launch MATLAB on the workstation and set the paths properly on MATLAB.
4. Launch CoppeliaSim on the workstation.
5. IMPORTANT: To facilitate the development of your forward kinematic model in this laboratory session, you should load the `Gen3Lite fk.ttt` scene, which contains the Gen3 lite manipulator model with no virtual objects, as shown in Figure 1. The scene file (.ttt file) is available in the \scenes directory of the simAPI path. Use the `loadscene()` command where appropriate in your command script.

```
v.loadscene('Gen3Lite fk','local',true)
```

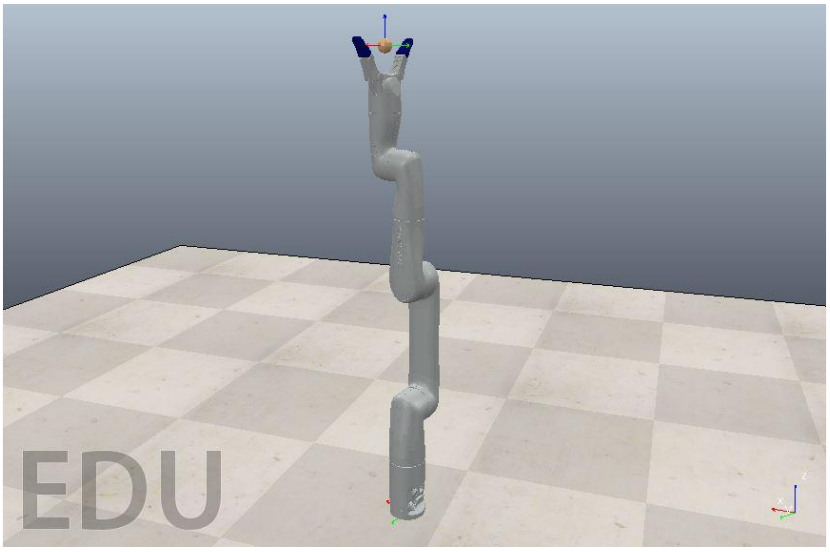


Figure 1: Scene with Gen3 lite model for forward kinematic modeling (`Gen3Lite fk.ttt`)

3.2 Observe and understand how the robot moves

1. Building upon your observations during Lab. #1, further inspect the direction of movement for every joint individually on the CoppeliaSim simulator when you increase (or decrease) the corresponding joint angle. Your objective is to determine what corresponds to the positive direction of rotation for every joint

of the Gen3 lite robot. This will constrain the positive direction of the *Z* rotation axes when setting the *n+1* reference frames along the structure of the robot in accordance with the Denavit-Hartenberg convention.

3.3 Develop the Gen3 lite robot's forward kinematic model

1. After you will have identified the location of all revolute joints along the robot arm structure along with their respective positive direction of rotation, use the two diagrams in Figure 20 on p. 53 of the user manual "*Gen3_lite_USER_GUIDE_release2021.pdf*" (also available in Figure 2 below) to assign all necessary *Z* and *X* axes on the manipulator, in accordance with the Denavit-Hartenberg convention.

   The configuration depicted in Figure 20 on p. 53 of the user manual represents the manipulator from two different viewpoints when the robot is in its "home" configuration (named "zero" by Kinova), that is when all joints have their values set to [0°, 0°, 0°, 0°, 0°, 0°]. Pay attention to the position of the electrical connectors illustrated at the base of the robot on the left image, and to the protruding section of the wrist toward the right on the right image, to help yourself visualize the two different viewpoints.

   The manufacturer also defines two fundamental reference frames that must be taken into account while developing your forward kinematic model. These are the "base" and the "tool" reference frames, with the latter located at the 2-finger gripper level. These reference frames can be observed on p. 42 of the user guide "*Gen3_lite_USER_GUIDE_release2021.pdf*", as well as on Figure 3 below.

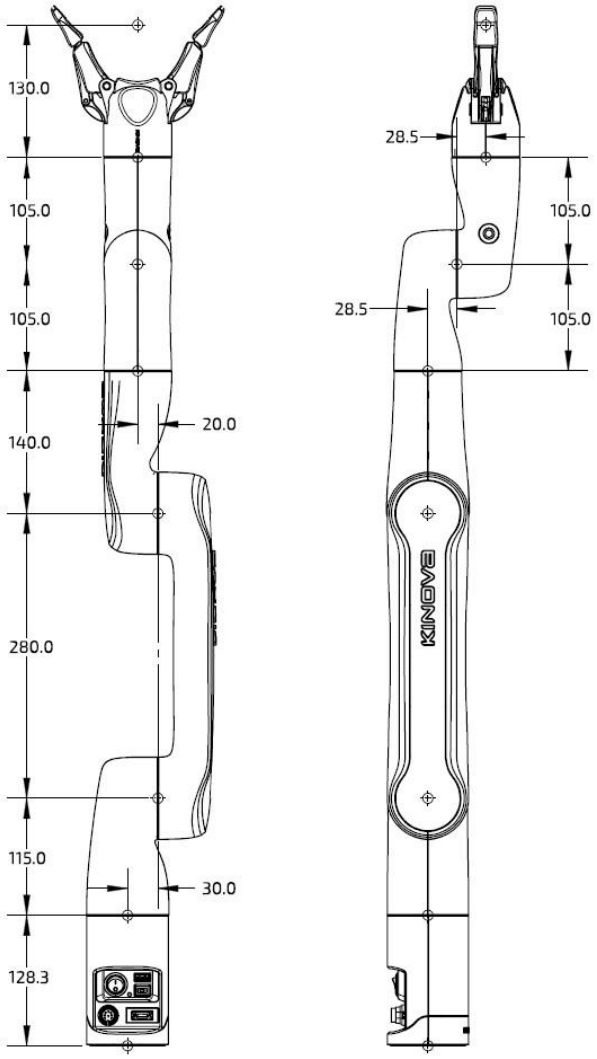


Figure 2: Views of Gen3 lite with joints set to [0°, 0°, 0°, 0°, 0°, 0°]

Table 1: Denavit-Hartenberg parameters table for the Gen3 lite manipulator.

Joint	l_i	d_i	α_i	θ_i
1				
2				
3				
4				
5				
6				

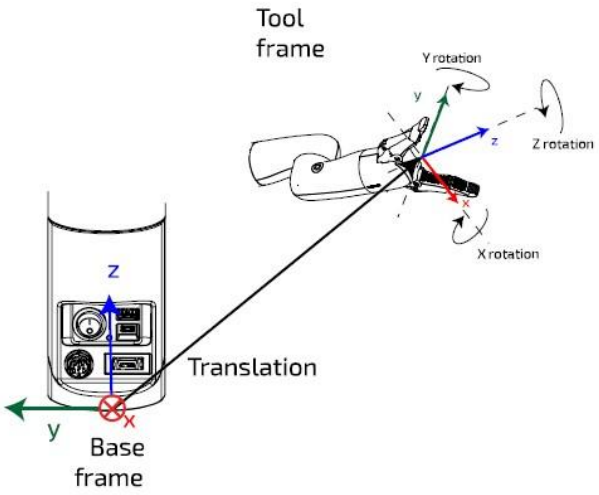


Figure 3: Details of the "base" and "tool" reference frames on the Gen3 lite robot

2. Considering all that information, your objective is to define the reference frames and the parameters of the robot that will accurately capture its behavior in any possible configuration, that is for any admissible set of the joint angular values. The observations that you make from the diagrams on Figure 2 above (or Figure 20 on p. 53 of the user guide) will influence the offsets needed to compensate on the joint variables [θ_1, θ_2, θ_3, θ_4, θ_5, θ_6] in the Denavit-Hartenberg table (Table 1). Do not hesitate to help yourself with the simulator if you wish to more closely observe the robot in other configurations while properly assigning the reference frames.

   For your resulting forward kinematic model to match with the representation of the robot implemented in CoppeliaSim, you should definitely adopt the definition of R_{base} and R_{tool} as shown on Figure 3. These are the two reference frames between which you must determine a forward transformation matrix, $Q_{EndEffector/base}$, where $R_{EndEffector} = R_{tool}$. The simulator will also report on 3D geometrical transformation parameters based on this pair of reference frames, and that must therefore be coherent with your mathematical model.

3. Next, examine your reference frames and fill in the Denavit-Hartenberg parameters in Table 1 for all six degrees of freedom. Use the lengths presented in Figure 2 above (or Figure 20 on p. 53 of the user guide "*Gen3_lite_USER_GUIDE_release2021.pdf*"). Relate these measurements to the sets of *X-Z* axes that you defined above and fill in the DH table for parameters l_i, d_i, α_i and θ_i. Do not forget to apply the proper compensation offsets on the θ_i parameters for the model to properly match with the home configuration (that is with the robot considered to be all straight up when all joints are set to 0°). Again, you may use the simulator to assign the robot to various configurations and validate your offsets.

You must figure out a proper formulation for the set of axes, the DH parameters and the overall forward kinematic model that will be coherent with what you observe on CoppeliaSim when loading the `Gen3Lite fk.ttt` scene and actuating its joints by sending commands that specify a variety of joint angular values.

4. Complete the development of the forward kinematic model by defining each one of the six *A* matrices and performing the product to obtain the theoretical $Q_{\text{EndEffector/base (DH model)}}$ transformation matrix. Preserve the angular parameters as variables for every degree of freedom [θ_1, θ_2, θ_3, θ_4, θ_5, θ_6] such that you can assign them any values later on. Implement your forward kinematic in MATLAB for this transformation matrix to be numerically estimated when you provide a specific set of joint angular values.
5. Investigate whether your forward kinematic model is accurate by selecting different sets of joint coordinates and assigning those sets simultaneously to your theoretical kinematic model and to the simulated Gen3 lite manipulator via CoppeliaSim. Begin with simple configurations such as [90°, 90°, 90°, 90°, 90°, 90°] or [-90°, -90°, -90°, -90°, -90°, -90°]. You may also validate with other reference configurations defined by Kinova, as depicted in Figure 4 below. Carefully validate the results. Use the knowledge acquired in Lab. #1 to operate the simulator via MATLAB.

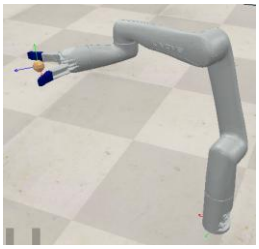
Configuration "home" *[θ_1, θ_2, θ_3, θ_4, θ_5, θ_6] = [0°, 344°, 75°, 0°, 300°, 0°]* *or with compensation for joint limits* *[θ_1, θ_2, θ_3, θ_4, θ_5, θ_6] = [0°, -16°, 75°, 0°, -60°, 0°]*	
Configuration "retract" *[θ_1, θ_2, θ_3, θ_4, θ_5, θ_6] = [357°, 21°, 150°, 272°, 320°, 273°]* *or with compensation for joint limits* *[θ_1, θ_2, θ_3, θ_4, θ_5, θ_6] = [-3°, 21°, 150°, -88°, -40°, -87°]*	
Configuration "packaging" *[θ_1, θ_2, θ_3, θ_4, θ_5, θ_6] = [270°, 148°, 148°, 270°, 140°, 0°]* *or with compensation for joint limits* *[θ_1, θ_2, θ_3, θ_4, θ_5, θ_6] = [-90°, 148°, 148°, -90°, 140°, 0°]*	

Figure 4: Various joint configurations of reference defined by the manufacturer for the Gen3 lite manipulator

After you have set selected values for the joint coordinates, examine the relative position and orientation of the end effector in the simulator. The end effector's pose is depicted by a reference frame attached to a yellow sphere in between the two fingers and is measured with respect to the base reference frame visible at the very bottom of the robot. Note that these frames will only be visible after the simulation has started (`v.simstart()`) and not immediately upon loading the scene `Gen3Lite fk.ttt`. Closely inspect how the *X-Y-Z* axes (represented respectively with Red-Green-Blue vectors) of both frames relate to each other, and compare with the interpretation of numerical values in your theoretical forward kinematic model matrix. You may use the "atan2" function extraction strategy discussed in the course to interpret the numerical values from your $Q_{\text{EndEffector/Base (DH model)}}$ transformation matrix, if needed.

You can also ask CoppeliaSim to output the six values that correspond to the geometrical degrees of freedom of the transformation matrix that represents the configuration of $R_{EndEffector}$ with respect to R_{base} in the current configuration of the manipulator. Call for the properties of the Gen3Lite instance "j" on MATLAB, by typing *j <enter>*, and it will output a list of parameters, including the current robot joints position "*JointPos*" (in radians), the current fingers position "*FingerPos*" (in % of closure), and the current end effector pose "*EndEffectorPose*" as shown in Figure 5.

```
j =

  GEN3LITE with properties:

       IsConnected: 1
          JointPos: [-0.0056 -0.2801 1.3089 -0.0088 -1.0452 -0.0033]
         FingerPos: 9.7441e-05
     EndEffectorPose: [0.4412 0.1908 0.4425 -1.6027 0.5326 -3.1088]
               DOF: 6
```

Figure 5: Extracting the current end effector's pose with respect to the base in CoppeliaSim

CoppeliaSim encodes Cartesian coordinates following the Tait-Bryan angles convention, which is a variation of Euler angles but with rotations considered around the three X,Y,Z axes. The Tait-Bryan convention encodes a 3D transformation defined as:

$$Q_{Tait-Bryan} = Q_T(x_T, y_T, z_T)\, Q_{Rx}(\alpha)\, Q_{Ry}(\beta)\, Q_{Rz}(\gamma)$$

where x_T, y_T, z_T are the first three parameters (in meters) respectively; α is the fourth, β the fifth, and γ the sixth parameters (in radians) respectively, all contained in the "*EndEffectorPose*" vector shown in Figure 5.

Using the six output parameters, calculate the $Q_{\text{EndEffector/Base (Simulator)}}$ transformation matrix that represents the current configuration of the simulated Gen3 lite robot, that is $R_{EndEffector}$ with respect to R_{base}, in order to compare it with the theoretical $Q_{\text{EndEffector/Base (DH model)}}$ transformation matrix

that you obtained for the same set of joint angles using the forward kinematic model that you developed.

3.4 Experimentally validate the forward kinematic model

Validate the correctness of your forward kinematic model by assigning a variety of joint configurations that will sample different regions of the robot workspace. For each imposed joint configuration, calculate the numerical representation $Q_{\text{EndEffector/Base (DH model)}}$ of the pose of the end effector (R_{tool}) with respect to R_{base} as defined in Section 3.3.

Also extract the numerical representation $Q_{\text{EndEffector/Base (Simulator)}}$ defining the current pose of the simulated end effector (R_{tool}) with respect to R_{base} in CoppeliaSim. The mathematical forward kinematic model should match with the simulated robot configuration as graphically depicted in the simulator and reported by the "*EndEffectorPose*" properties of the Gen3Lite instance "j". Make sure that the two corresponding transformation matrices coincide. The diagram in Figure 6 illustrates what you are aiming for, over a diversity of robot's configuration. If the two transformation matrices do not match, revise your forward kinematic model until it accurately represents the proper position and orientation to match the simulated robot.

Hint: If you notice substantial discrepancies between the $Q_{\text{EndEffector/Base (DH model)}}$ and the $Q_{\text{EndEffector/Base (Simulator)}}$ numerical representations while the joint values are not completely off, the simulator may have cumulated errors due to consideration of dynamic effects. Refresh the simulator by performing a clean disconnect, stop simulation, and clear scene; and then reload the `Gen3Lite fk.ttt` scene, restart simulation and reconnect to a new Gen3Lite instance. Then test your kinematic model again on the fresh version of the scene in the simulator. If significant discrepancies are still visible, your model is probably inadequate.

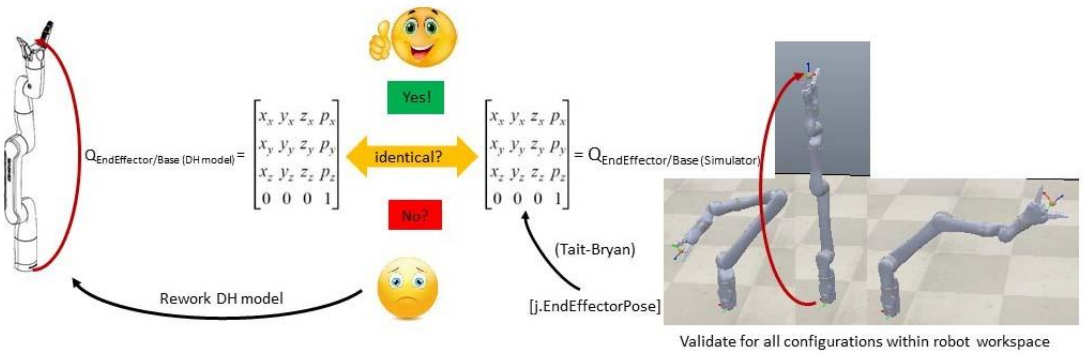


Figure 6: Validating the correctness of forward kinematic model against the simulator

4 Laboratory Experimental Activities Evaluation

To complete this second laboratory session, proceed with the online evaluation that is proposed on Brightspace, under Quizzes, "Lab 2 Evaluation". This evaluation must be performed **by team** (only one submission per team, indicate the names of all team members) and **submitted at the latest by Friday, Nov. 6th, 2026 at 11:59PM**.

You will be asked to report on the parameters (axes, DH table) of your forward kinematic model (provide a copy of page 4 of this document that you will have filled) and to demonstrate its accuracy over a variety of randomly assigned joint configurations that will be provided to you when you will complete the online evaluation.

Have your forward kinematic model running on MATLAB as well as your MATLAB script ready to control the simulated Gen3 lite robot on CoppeliaSim. Then, access Brightspace and complete the "**Lab 2 Evaluation**" quiz.

When you will be asked to provide a screenshot image of the Gen3 lite manipulator in the imposed configuration, capture a screenshot image of only the area of the manipulator included in the `Gen3Lite fk.ttt` scene. Privilege the default viewpoint of the virtual observer in CoppeliaSim to facilitate comparison between configurations. Figure 7 shows a sample of an expected screenshot image of the robot in its default configuration, with all joints set to 0°. Screenshot samples of alternative configurations are also shown in Figure 4.



Figure 7: A sample screenshot image of the Gen3 lite manipulator in its home configuration

There is no formal lab report to submit for this lab session beyond what is asked for on Brightspace.

5 Disconnect from Laboratory

Safely close your simulation as instructed in Lab. #1 guide.